Data Communication Engineering

Assignment 1

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Investigate and analyze the packets by using WireShark when executing two commands, ping and traceroute for the following two network connections and two destination hosts (totally four conditions). 1) Compare ARP and ICMP packets with the packet format described in the slides, and 2) explain how ARP and ICMP packets are sent and received by your computer with your sentences, screenshots, and timeline flowcharts. Also, 3) consider and discuss the differences among the conditions.

- Applications you try to use:
  (1) ping
  (2) traceroute

- Network connections (Do not use VPN)
  (1) PC connected to SRAS Wi-Fi or PC connected to your Laboratory
  (2) PC in your house or PC connected to eduroam

- Destination hosts
  (1) www.yamazaki.se.shibaura-it.ac.jp (Prof. Yamazaki Lab. web site)
  (2) www.google.co.jp (Google web site)

Due date: Before the next class (9:00 Nov. 9, 2026)

Submission format: PowerPoint PPT file (or PDF)